EE2002-Tutorial 3 Solutions

1. Simplify the Boolean expressions using 3-variable K-maps.

(a) $F(x, y, z) = (3, 4, 6, 7)$

(b) $F(x, y, z) = (0, 2, 4, 5, 6, 7)$

(c) $F(x, y, z) = (1, 3, 5, 7, 6)$

Solution:

(a) $F(x, y, z) = (3, 4, 6, 7)$

x \ yz	00	01	11	10
0	0	0	1	0
1	1	0	1	1

Figure 1: K-map for $F(x, y, z) = (3, 4, 6, 7)$.

1. Populate the K-map using the given minterms: $3, 4, 6, 7$.
2. Group adjacent 1s into the largest possible power-of-2 groups.
3. Write down the simplified Boolean expression for each group:
   - Group 1: Corresponds to yz.
   - Group 2: Corresponds to xz'.
4. Combine the simplified expressions to obtain the final result:

$$F = yz + xz'$$

Solution:

(b) $F(x, y, z) = (0, 2, 4, 5, 6, 7)$

1. Populate the K-map using the given minterms: $0, 2, 4, 5, 6, 7$.
2. Group adjacent 1s into the largest possible power-of-2 groups.

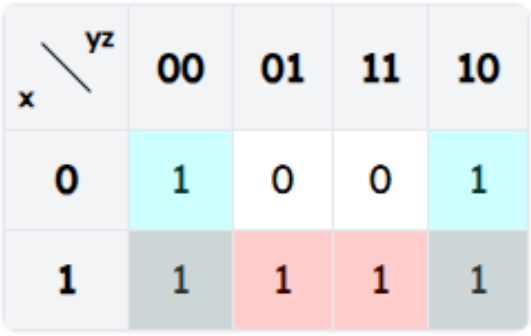


Figure 2: K-map for $F(x, y, z) = (0, 2, 4, 5, 6, 7)$.

3. In this K-map, we can group two quads:
   - Group 1: Corresponds to x.
   - Group 2: Corresponds to z'.
4. Combine the simplified expressions to obtain the final result:

$$F = x + z'$$

Solution:

(c) $F(x, y, z) = (1, 3, 5, 7, 6)$

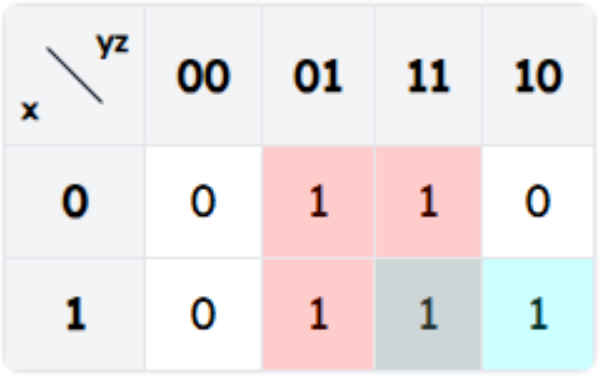


Figure 3: K-map for $F(x, y, z) = (1, 3, 5, 7, 6)$.

1. Populate the K-map using the given minterms: $1, 3, 5, 7, 6$.
2. Group adjacent 1s into the largest possible power-of-2 groups.
3. In this K-map, we can group one quad and one pair as shown in the figure.
4. Write down the simplified Boolean expression for each group:
   - Group 1 (Quad): Corresponds to z.

- Group 2 (Pair): Corresponds to xy.

5. Combine the simplified expressions to obtain the final result:

$$F = z + xy$$

2. Simplify the following Boolean function using a Karnaugh map (K-map) and identify the prime implicants and essential prime implicants:

$$Y = \sum(0, 2, 3, 4, 9, 10, 12, 13) + d(6, 8, 14)$$

Solution: The K-Map is :

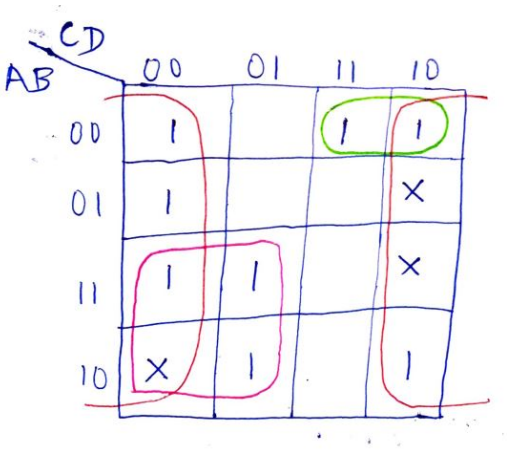


Figure 4: K-map for $Y = \sum(0, 2, 3, 4, 9, 10, 12, 13) + d(6, 8, 14)$.

1. Populate the K-map using the given minterms and don't-care terms:
   - Minterms: $0, 2, 3, 4, 9, 10, 12, 13$.
   - Don't-care terms: $6, 8, 14$.
2. Group adjacent 1s and don't-care terms into the largest possible power-of-2 groups.
3. Identify the prime implicants.
   - The prime implicants are: $D', AC', A'B'C$.
4. Identify the essential prime implicants.
   - The essential prime implicants are: $D', AC', A'B'C$.
5. The simplified Boolean expression:

$$Y = D' + AC' + A'B'C$$

3. A logic circuit realizes the function $F(a,b,c,d) = a'b' + a'cd + ac'd + ab'd'$. Assuming that $a = c$ never occurs when $b = d = 1$, find a simplified expression for F.

Solution:

$$F(a,b,c,d) = a'b' + a'cd + ac'd + ab'd'$$

From the given Boolean function, the minterms for the function are:

$$\sum m(0,1,2,3,7,8,9,10,13)$$

Step 1: Focus on Rows Where $b = 1$ and $d = 1$

We focus on the rows where both $b = 1$ and $d = 1$, because this is where the condition $a = c$ never happens. The relevant rows in the truth table are:

a	b	c	d	$F(a,b,c,d)$
0	1	0	1	X
1	1	1	1	X

From the above table, it is evident that the minterms $m(5)$ and $m(15)$ are don't care conditions.

Step 2: Treat Minterms 5 and 15 as Don't Care

Since minterms $m(5)$ and $m(15)$ are don't care conditions, we treat them as flexible. These don't care conditions do not affect the simplification and can be considered either 1 or 0 depending on what helps us minimize the Boolean expression.

ab \ cd	00	01	11	10
00	1	1	1	1
01	0	X	1	0
11	0	1	X	0
10	1	1	0	1

Figure 5: K-map

Step 3: Minimize Using a Karnaugh Map (K-map). The K-map is shown in Fig. 5.

The minimised expression in SOP form:

$$F(a, b, c, d) = c'd + a'd + b'd'$$

4. Consider the Boolean function $z(a, b, c)$ in Fig. 6. Which of the following minterms represent this circuit?

(a) $z = \Sigma(0, 1, 3, 7)$

(b) $z = \Sigma(1, 4, 5, 6, 7)$

(c) $z = \Sigma(2, 4, 5, 6, 7)$

(d) $z = \Sigma(2, 3, 5)$

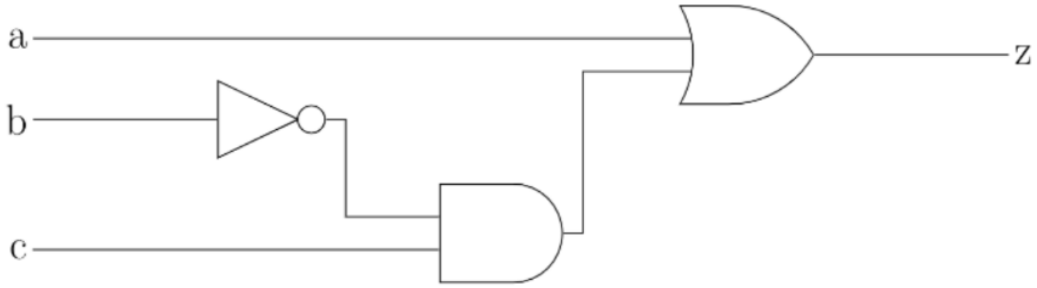


Figure 6:

Solution:

1. From the given Boolean circuit, the simplified Boolean expression is:

$$z = a + b'c$$

2. To find the minterms, we need to evaluate the circuit for all possible combinations of a, b, and c.

3. Evaluate the expression $z = a + b'c$ for each combination of a, b, and c:

a	b	c	z
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

4. From the above evaluations, the minterms where $z = 1$ are: $1, 4, 5, 6, 7$.

5. Thus, the minterms for the given Boolean function are:

$$z = \Sigma(1, 4, 5, 6, 7)$$

5. A plot of minterms is given in Figure 7. Draw the K-map and simplify.

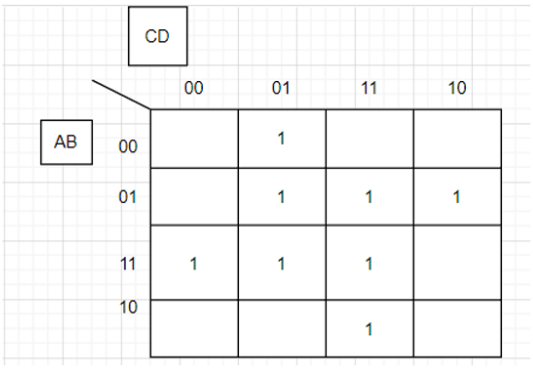


Figure 7: Plot of minterms

Solution:

1. Group adjacent 1s into the largest possible power-of-2 groups.
2. Identify the essential prime implicants by examining each group:
   - Group 1: Corresponds to $A'C'D$.
   - Group 2: Corresponds to $A'BC$.
   - Group 3: Corresponds to ACD.
   - Group 4: Corresponds to ABC'.
3. The K-map solution is shown in the figure below:

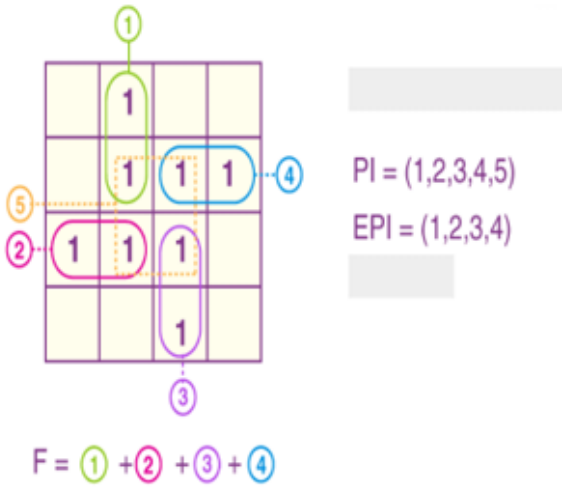


Figure 8: K-map solution for the simplified expression.

4. Combine the prime implicants to obtain the simplified Boolean expression:

$$f(A, B, C, D) = A'C'D + A'BC + ACD + ABC'$$

6. A logic circuit realizing a function f has four inputs *a, b, c* and *d*. The three inputs *a, b*, and *c* are the binary representations of the digits 0 through 7 with *a* being the most significant bit. The input *d* is an odd-parity bit; that is, the value of *d* is such that *a, b, c,* and *d* always contain an odd number of 1s. (For example, the digit 1 is represented by abc = 001 and d = 0, and the digit 3 is represented by abcd = 0111.) The function f has value 1 if the input digit is a prime number. Cells which correspond to an even number of 1s represent don't cares. Draw a Karnaugh map for f and simplify to get all minimum sum of products for f.

Solution:

We make a few observations.

1. The inputs to the logic circuit are four variables: a, b, c, and d. The first three variables a, b, and c represent a 3-bit binary number that can take values from 0 to 7. The variable d is the odd-parity bit such that the total number of 1s in the 4 bits a, b, c, d is always odd.

2. Note that the parity bit d is determined based on the number of 1s in the 3-bit binary number formed by a, b, c:

- If the number of 1s in a, b, c is odd, then $d = 0$.
- If the number of 1s in a, b, c is even, then $d = 1$.

3. The output function f is 1 if the number corresponding to a, b, c is a prime number. Otherwise, $f = 0$. The prime numbers between 0 and 7 are: 2, 3, 5, and 7. Therefore, the function f is defined as:

$$f(a,b,c) = \begin{cases} 1 & \text{if the decimal equivalent of } abc \text{ is a prime number} \\ 0 & \text{otherwise} \end{cases}$$

4. Don't Care Cases: When the number of 1s in the bits a, b, c, d is even, the corresponding cell is marked as a don't care (X).

5. The Karnaugh map is shown in Figure 9.

The minimised SOP expression is :

$$F(a,b,c,d) = a'b + ac$$

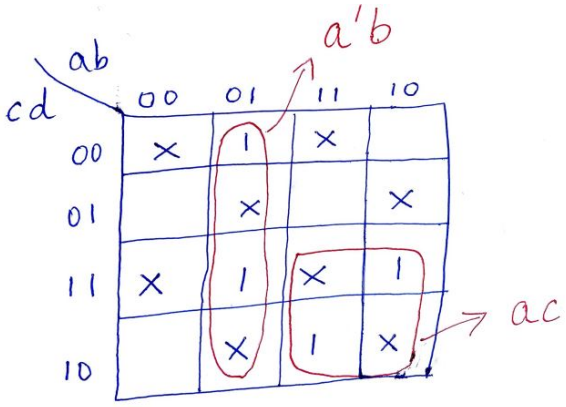


Figure 9: K-Map

7. The plot of minterms for a four-variable Boolean function is given in Fig. 10. Find the minimal POS form using this K-map.



Figure 10:

Solution: The K-map for a four-variable Boolean function is shown in Fig. 11.

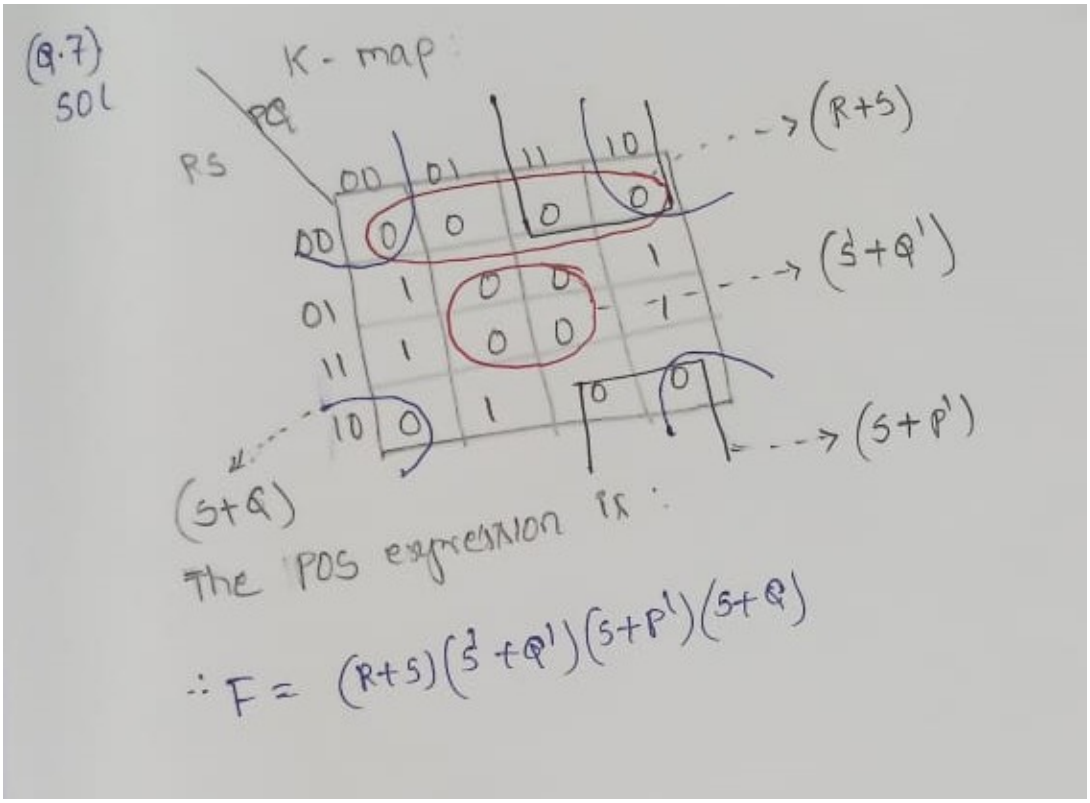


Figure 11: K-map for the four-variable Boolean function.

1. Identify the cells with a value of 0 in the K-map.
2. Group these cells into the largest possible power-of-2 groups.
3. Write down the product terms (OR terms) for each group by identifying the variables that remain constant in the 0-groups:
   - Group 1: Corresponds to $(R + S)$.
   - Group 2: Corresponds to $(S' + Q')$.
   - Group 3: Corresponds to $(S + P')$.
   - Group 4: Corresponds to $(S + Q)$.
4. Combine all the product terms using AND to obtain the minimal POS form.

Minimal POS Form:

$$F = (R + S)(S' + Q')(S + P')(S + Q)$$

8. For $F(a, b, c, d) = \pi(0, 1, 3, 7)$, the minimum number of terms in the minimal SOP form is given by 'X'. Find the value of 'X'. Explain your solution.

Solution: The K-map is shown below

AB \ CD	00	01	11	10
00	0	0	0	1
01	1	1	0	1
11	1	1	1	1
10	1	1	1	1

Figure 12: Solution for the given Boolean function.

Steps to Determine the Minimal SOP Form:

1. The given function is represented in the canonical POS form: $\pi(0, 1, 3, 7)$, which corresponds to the minterms where $F = 0$.
2. Convert the POS form to the SOP form by identifying the maxterms $(0, 1, 3, 7)$ as the cells where $F = 0$ in the K-map.
3. Group adjacent 1s (not 0s) in the K-map to form the largest possible power-of-2 groups.
4. Write down the minimal SOP expression based on the grouping:
$$F = A + BC' + CD'$$
5. Count the number of terms in the minimal SOP form. Here, the number of terms is $X = 3$.

9. Draw a logic diagram using only two-input NOR gates to implement the following:

$$F(A, B, C, D) = (A \oplus B)'(C \oplus D)$$

Solution:

1. The XOR operation $A \oplus B$ can be written as:
$$A \oplus B = AB' + A'B$$

Similarly, $C \oplus D$ is:

$$C \oplus D = CD' + C'D$$

Therefore, the expression becomes:

$$F(A, B, C, D) = (AB + A'B')(CD' + C'D)$$

$$F(A, B, C, D) = ABCD' + ABC'D + A'B'CD' + A'B'C'D$$

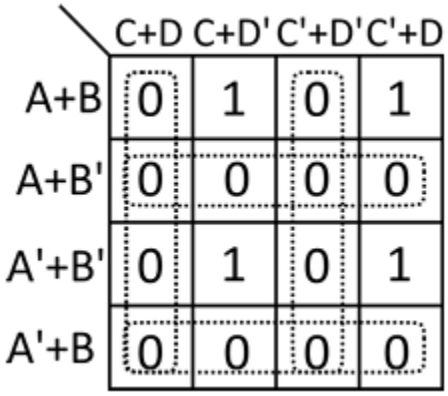


Figure 13: K-map for the expression $F = (A \oplus B)'(C \oplus D)$.

2. From K-map the expression $F(A, B, C, D)$ using NOR gates is:

$$F = (A + B')(A' + B)(C + D)(C' + D')$$

$$F = \left[((A + B')(A' + B))' + ((C + D)(C' + D'))'\right]'$$

$$F = \left[\left[((A + B')' + (A' + B)')'\right]' + \left[((C + D)' + (C' + D')')'\right]'\right]'$$

3. The logic diagram using two-input NOR gates is shown in the following figures:

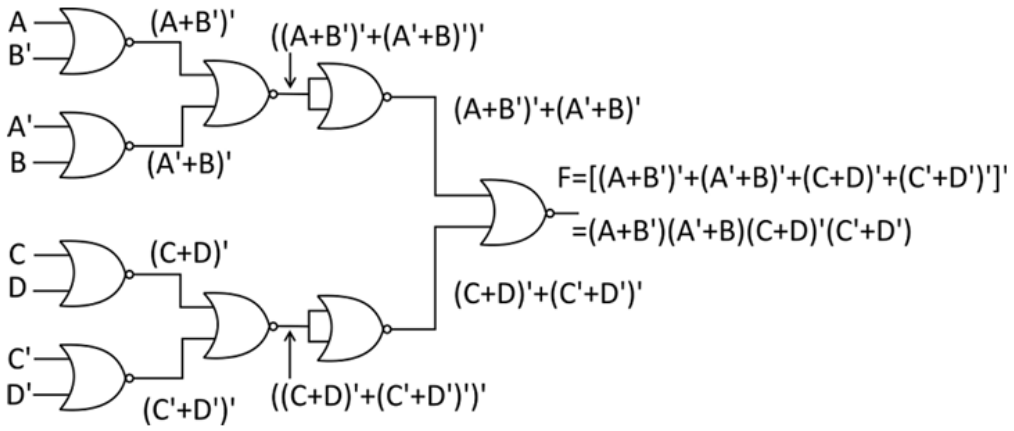
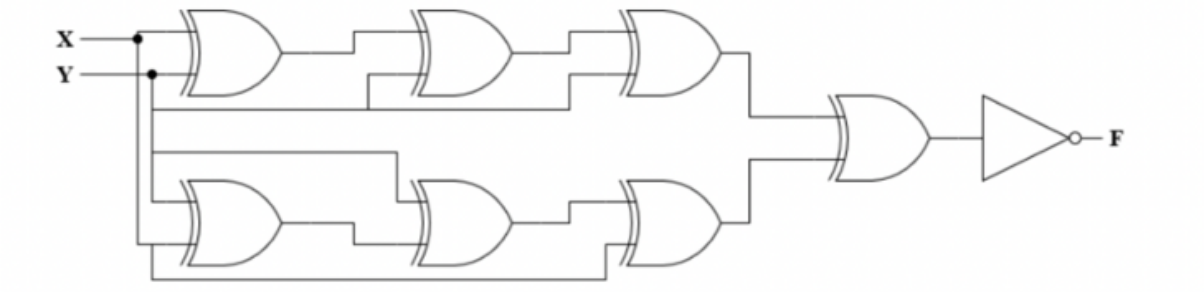


Figure 14: Logic diagram using two-input NOR gates.

10. What is the output of the circuit shown below?



Solution: The function represented by the circuit, namely F, can be expressed as

$$F = \overline{F_1 \oplus F_2}$$

where F_1 and F_2 are as follows:

$$F_1 = x \oplus y \oplus y \oplus y$$

Since

$$y \oplus y = 0$$

$$F_1 = x \oplus y \oplus 0$$

Further, since $(y \oplus 0 = y)$, we have

$$F_1 = x \oplus y$$

Similarly, F_2 can be expressed as follows:

$$F_2 = x \oplus y \oplus y \oplus x$$

Using properties of XOR, namely

$$x \oplus x = 0 \quad \text{and} \quad y \oplus y = 0$$

we have

$$F_2 = 0$$

Step 3: Combine F_1 and F_2:

$$F = \overline{F_1 \oplus F_2}$$

Substituting the values of F_1 and F_2, we have:

$$F = \overline{(x \oplus y) \oplus 0}$$

$$F = \overline{x \oplus y} = x \odot y$$